

Aditya Karnam Gururaj Rao

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 12+ years of experience across AI/ML, distributed systems, and scalable engineering in ad-tech, fintech, blockchain, and SaaS. Tech Lead at Zefr, currently building **ZAiN**, a data-driven AI agent, covering agent architecture, MCP servers, tools, evals, agent gateway, permissioning, and observability, alongside customer-facing platform systems on AWS and GCP. Published AI researcher (IEEE, Frontiers, arXiv) with work spanning medical LLMs, healthcare AI, and large-scale community detection, and open-source maintainer with contributions accepted by Anthropic. Delivered \$350K+ in direct business value, including Meta measurement systems with 83% performance gains, vector search infrastructure over 100M records, and two company hackathon wins shipped to production. Previously founded and scaled a SaaS platform serving 100+ clients with a team of 10+ developers.

EMPLOYMENT HISTORY

ZEFR Jan 2022 – Present
Los Angeles, California, United States, promoted from Software Engineer to Senior Software Engineer across four levels

Senior Software Engineer, Tech Lead of Customer Experience (Dec 2025 – Present)

- ◆ Building **ZAiN**, Zefr's data-driven AI agent: architecting agents, MCP servers, tools, evals, agent gateway, permissioning, session compaction, and chat interface design, including MCP apps for Claude Code, Codex, and Microsoft Copilot 365.
- ◆ Build observability and monitoring around agent logs for production AI behavior.
- ◆ Drive architecture and long-term technical direction for customer-facing and internal platform systems across AWS and GCP.
- ◆ Lead distributed system design spanning Kafka-based campaign processing, Airflow orchestration, search infrastructure, API gateway architecture (Apigee/n8n), and AI integrations, while delivering scalable reporting and data solutions.
- ◆ Author RFCs, lead design reviews, mentor engineers, and influence cross-team technical decisions to raise engineering standards and platform reliability.

Software Engineer to Software Engineer III (Jan 2022 – Dec 2025)

- ◆ Worked with Python, Scala, Elasticsearch, Kafka, SQL, Shell script, Docker, Git; AWS ECS, EC2, Firehose, S3, Lambda, Redshift, Postgres, CloudWatch.
- ◆ Developed, optimized, deployed, and maintained the inventory availability endpoint for digital property.
- ◆ Used LaunchDarkly feature flags to control Elasticsearch filters across environments.
- ◆ Listened to, processed, and published data to multiple Kafka topics for internal messaging.
- ◆ Developed a tool to upload custom Redshift metrics to AWS CloudWatch and Datadog.

DIGITAL SYSTEMS ARCHITECT

Jun 2021 – Jan 2022

ePlata, Edwardsville, Illinois, United States

- ◆ Built NFT platform on Cardano (Plutus Smart Contracts).
- ◆ Architected KYC Remittance API integrating Western Union APIs.
- ◆ Spearheaded IPC-V remittance platform generating \$3M transaction volume.
- ◆ Delivered full-stack engineering for 8+ white-label fintech clients (React, TypeScript, GraphQL, Docker).

FULL STACK ENGINEER

Feb 2020 – May 2021

ePlata, Edwardsville, Illinois, United States

- ◆ Migrated AWS EC2 workloads to ECS to optimize resource utilization and reduce costs.
- ◆ Mentored interns and built GraphQL API testing templates.
- ◆ Rebuilt logging service with support for Sentry, Logzio + PII sanitization.

SOFTWARE ENGINEER

Jun 2018 – Jan 2020

ePlata, United States

- ◆ Integrated Splunk ML Toolkit for Payment Fraud System.
- ◆ Improved build pipelines with AWS CodeBuild (75% time reduction).
- ◆ Built native contacts integration (Swift, Kotlin, TypeScript) increasing load performance by 60%.
- ◆ Created CI/CD pipelines with Docker, Jenkins, CircleCI, AWS CodeDeploy.

GRADUATE RESEARCH ASSISTANT

Aug 2017 – Jun 2018

University of Missouri–St. Louis, St. Louis, Missouri, United States

- ◆ Thesis: *Efficient Reduced-Bias Genetic Algorithm for Generic Community Detection Objectives*, a novel genetic algorithm

- framework for clustering networks with 5M+ nodes.
- ◆ Collaborated with Oak Ridge National Lab, using NVIDIA Titan cluster computing resources on an Alzheimer's gene-expression network dataset.
- ◆ Reduced memory usage with a 3-dimensional bit-array container.

FOUNDER & CTO

Aug 2014 – Aug 2016

eventzé

- ◆ Founded a SaaS platform for managing large college events covering registrations, merchandise, finance automation, and digital branding, partnering with 100+ college fests.
- ◆ Built and led a team of 10+ developers to scale the platform (PHP/CodeIgniter, ReactJS, REST APIs, payment gateway integration, DevOps automation).

SELECTED PROJECTS & OPEN SOURCE

- ◆ **quecto**: a tiny OpenAI-compatible AI harness and coding agent in Rust with a ~1.3 MB core and ~3.5 MB agent binary.
- ◆ **embenx**: a universal embedding retrieval toolkit and agentic memory layer with a unified API for 15+ vector backends and MCP support for Claude and autonomous agents (`pip install embenx`).
- ◆ **Portfolio MCP**: a public read-only Model Context Protocol server exposing portfolio projects, research, and source links as structured data for agents (adityakarnam.com/mcp-install).
- ◆ **locobench**: local coding model benchmarking, a Harbor and Ollama setup for running local agentic benchmarks with a leaderboard pipeline.
- ◆ **mcp-scholarly**: academic retrieval with MCP for scholarly search from AI assistants.
- ◆ **opencontext**: interoperable AI chat contexts across assistants (open-context.dev).
- ◆ **medfit-llm**: applied model fine-tuning in healthcare.
- ◆ **google-drive-upload-git-action**: developer productivity automation as a reusable GitHub Action.

EDUCATION

MS, Global Business and Blockchain , University of the Cumberland	2021 – 2022
MS, Computer Science , University of Missouri–St. Louis	2016 – 2018
B.Tech, Computer Science , Vellore Institute of Technology	2012 – 2016

PUBLICATIONS & RESEARCH

- ◆ *MEDFIT-LLM: Medical Enhancements through Domain-Focused Fine Tuning of Small Language Models*. IEEE, 2025.
- ◆ *CAG: Chunked Augmented Generation for Google Chrome's Built-in Gemini Nano*. arXiv:2412.18708, 2024.
- ◆ *AI-Enhanced Neural Network Framework for Cardiac Arrest Prognostics*. Frontiers in Health Informatics, 2024.
- ◆ *Automated Analysis of Glucose Response Patterns in Type 1 Diabetes Using Machine Learning and Computer Vision*. International Journal of Advanced Computer Science & Applications, 2025.
- ◆ *Neural Network-Enabled Diagnostic Framework for Correlating Ayurvedic Breathwork Protocols with Adaptive Skin Microbiome Dynamics*. Mathematical Modelling of Engineering Problems, 2025.
- ◆ *The Harness Engineer*. 2025.
- ◆ *Quantifying Adaptive Governance Success: A Mixed-Methods Analysis of 500 Organizations*. 2025.
- ◆ *Efficient Reduced-Bias Genetic Algorithm (ERBGA) for Generic Community Detection Objectives*. MS thesis, University of Missouri–St. Louis, 2018.
- ◆ *Transmission in Benes Network*. International Journal of Pure and Applied Mathematics, 2016.